

AIM:

To develop an Android application that allows users to enter data and store it in the device's external storage (SD card).

ALGORITHM:

1. Open Android Studio and create a new project using Empty Views Activity
2. Design the user interface using:
 - a. EditText for user input
 - b. Button to save data
3. Add required permission in AndroidManifest.xml: WRITE_EXTERNAL_STORAGE
4. Open the main activity file
5. Link UI components using findViewById()
6. Use MediaStore API to save data into external storage
7. Handle empty input cases.
8. Display confirmation using Toast message.
9. Run the application and verify file creation.

CODE:**XML Layout (activity_main.xml)**

XML

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:orientation="vertical"
```

```
    android:padding="20dp"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent">
```

```
<EditText
```

```
    android:id="@+id/inputText"
```

```
    android:hint="Enter data"
```

```
        android:layout_width="match_parent"

        android:layout_height="wrap_content"/>

<Button

    android:id="@+id/saveBtn"

    android:text="Save to SD Card"

    android:layout_width="match_parent"

    android:layout_height="wrap_content"/>

</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.experiment_11;

import android.content.ContentValues;
import android.os.Bundle;
import android.os.Environment;
import android.provider.MediaStore;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.io.OutputStream;

public class MainActivity extends AppCompatActivity {

    EditText inputText;

    Button saveBtn;

    @Override
```

```

protected void onCreate(Bundle savedInstanceState) {

    super.onCreate(savedInstanceState);

    setContentView(R.layout.activity_main);


    inputText = findViewById(R.id.inputText);

    saveBtn = findViewById(R.id.saveBtn);


    saveBtn.setOnClickListener(v -> {

        String data = inputText.getText().toString();

        if (data.isEmpty()) {

            Toast.makeText(this, "Enter some data", Toast.LENGTH_SHORT).show();

        } else {

            saveToDownloads(data);

        }

    });

}

```

```

private void saveToDownloads(String data) {

    try {

        String fileName = "mydata_" + System.currentTimeMillis() + ".txt";


        ContentValues values = new ContentValues();

        values.put(MediaStore.MediaColumns.DISPLAY_NAME, fileName);

        values.put(MediaStore.MediaColumns.MIME_TYPE, "text/plain");

        values.put(MediaStore.MediaColumns.RELATIVE_PATH,
Environment.DIRECTORY_DOWNLOADS);


        // Use ContentResolver to insert the file

        android.content.ContentResolver resolver = getContentResolver();

        android.net.Uri uri = resolver.insert(

```

```
MediaStore.Files.getContentUri("external"), values);

if (uri != null) {

    try (OutputStream os = resolver.openOutputStream(uri)) {

        os.write(data.getBytes());

        Toast.makeText(this, "Saved to Downloads!", Toast.LENGTH_LONG).show();

    }

} else {

    Toast.makeText(this, "Failed to save file", Toast.LENGTH_SHORT).show();

}

} catch (Exception e) {

    Toast.makeText(this, "Error saving file", Toast.LENGTH_SHORT).show();

    e.printStackTrace();

}

}

}
```

OUTPUT:



RESULT:

Thus, an Android application was successfully developed to store user input data in the device's external storage (SD card).